

# BioSpecInfo

## Software Bill of Materials (SBOM)

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Versione **bsi-v189** · commit `40c00f5`

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# Software Bill of Materials (SBOM)

BioSpecInfo · document generated by `tools/genera-sbom.js`.

|                         |                                                        |
|-------------------------|--------------------------------------------------------|
| Application version     | bsi-v189                                               |
| Commit                  | 40c00f56e8cd3c0387ef2a0851d61c7f919eb2d0               |
| Generated (UTC)         | 2026-10-07T16:54:05.524Z                               |
| Machine-readable format | <code>../evidence/sbom.cdx.json</code> — CycloneDX 1.5 |

The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

## 1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

| Component                                                                 | Version/File                      | Licence      | Size    | Role                                                                 |
|---------------------------------------------------------------------------|-----------------------------------|--------------|---------|----------------------------------------------------------------------|
| <b>RDKit MinimalLib (JS glue)</b> <br><sub>RDKit / Greg Landrum</sub>     | <code>RDKit_minimal.js</code>     | BSD-3-Clause | 125 kB  | Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing. |
| <b>RDKit MinimalLib (WebAssembly)</b> <br><sub>RDKit / Greg Landrum</sub> | <code>RDKit_minimal.wasm</code>   | BSD-3-Clause | 6753 kB | WebAssembly binary of the cheminformatics engine.                    |
| <b>three.js</b> <br><sub>mrdoob e collaboratori</sub>                     | <code>three.min.js</code>         | MIT          | 589 kB  | WebGL rendering engine for the 3D scenes.                            |
| <b>three.js — UnrealBloomPass</b> <br><sub>mrdoob e collaboratori</sub>   | <code>three_bloom.js</code>       | MIT          | 25 kB   | Post-processing effect for the astronomical visualisations.          |
| <b>three.js — GLTFLoader</b> <br><sub>mrdoob e collaboratori</sub>        | <code>gltf_loader.js</code>       | MIT          | 94 kB   | Loading of anatomical and molecular models in glTF format.           |
| <b>3Dmol.js</b> <br><sub>University of Pittsburgh / David Koes</sub>      | <code>3Dmol-min.js</code>         | BSD-3-Clause | 502 kB  | 3D molecular visualisation (PDB, SDF, surfaces).                     |
| <b>SmilesDrawer</b> <br><sub>Daniel Probst</sub>                          | <code>smiles-drawer.min.js</code> | MIT          | 246 kB  | 2D structure drawing from SMILES.                                    |
| <b>sql.js (JS glue)</b> <br><sub>sql.js / SQLite</sub>                    | <code>lib/sql-wasm.js</code>      | MIT          | 45 kB   | In-browser SQLite database for the data exercises.                   |

| Component                                                                    | Version/File      | Licence       | Size   | Role                                                 |
|------------------------------------------------------------------------------|-------------------|---------------|--------|------------------------------------------------------|
| <b>SQLite (WebAssembly)</b> <br><sub>SQLite Consortium</sub>                 | lib/sql-wasm.wasm | Public Domain | 644 kB | SQLite engine compiled to WebAssembly.               |
| <b>Dataset dimuoni (CERN Open Data)</b> <br><sub>CERN Open Data Portal</sub> | lib/dimuon.js     | CC0-1.0       | 48 kB  | Experimental data for the particle-physics exercise. |

## Digests

| File                 | SHA-256                                                          |
|----------------------|------------------------------------------------------------------|
| RDKit_minimal.js     | 58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9 |
| RDKit_minimal.wasm   | e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8 |
| three.min.js         | 9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2 |
| three_bloom.js       | 42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090 |
| glTF_loader.js       | 5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6 |
| 3Dmol-min.js         | 06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff |
| smiles-drawer.min.js | 955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2 |
| lib/sql-wasm.js      | 77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f |
| lib/sql-wasm.wasm    | 438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed |
| lib/dimuon.js        | 61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d |

## 2. Code written for this project

| File            | Size    | Role                                                                                                                                                                                                                                                             |
|-----------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| index.html      | 4690 kB | Main application: interface, teaching sections, chemical data.                                                                                                                                                                                                   |
| bsi-ai-hub.js   | 422 kB  | «Spectra» AI agent: agentic loop, tools, provider management.                                                                                                                                                                                                    |
| bsi-spettri.js  | 35 kB   | IR/NMR spectral prediction engine over the molecular graph.                                                                                                                                                                                                      |
| bsi-cheminfo.js | 125 kB  | Cheminformatics engine: standardisation, fingerprints, clustering, PCA, QSAR models with a null model.                                                                                                                                                           |
| bsi-prettsch.js | 37 kB   | The NMR estimation tables transcribed in full from Prettsch-Bühlmann-Badertscher, 4th ed.: <sup>13</sup> C and <sup>1</sup> H benzenes, ethylenes, alkanes, alkynes, <sup>13</sup> C aliphatics, steric corrections, <sup>1</sup> J(C,H). Numbers only, no code. |
| bsi-nmr.js      | 56 kB   | Per-atom <sup>1</sup> H and <sup>13</sup> C NMR prediction: applies the bsi-prettsch.js tables by walking the ring and measuring distances, with a cyclic reference compound and chemical equivalence by environment code.                                       |
| bsi-moldraw.js  | 29 kB   | Interactive NMR spectrum: zoom, assignment table, peak↔atom link, CSV/JCAMP-DX/PNG exports.                                                                                                                                                                      |

| File                   | Size    | Role                                                                                                                |
|------------------------|---------|---------------------------------------------------------------------------------------------------------------------|
| bsi-spettrolettore.js  | 53 kB   | Spectrum reader: JCAMP-DX with ASDF compression, prominence-based peak finding, IR band assignment, neutral losses. |
| bsi-mol3d.js           | 41 kB   | 3D molecule viewer: formation animation, atom selection, measurement of angles, lengths and dihedrals.              |
| bsi-lingue.js          | 159 kB  | Internationalisation engine: fourteen languages, writing direction included.                                        |
| bsi-molingue.js        | 54 kB   | Chemical languages: twenty-five outputs from SMILES, with IUPAC naming over a declared class.                       |
| bsi-pannelli-lingua.js | 66 kB   | Panels for languages and chemical languages.                                                                        |
| sw.js                  | 5 kB    | Service Worker: offline operation and network strategy.                                                             |
| rdkit_lab.html         | 260 kB  | Cheminformatics laboratory.                                                                                         |
| astro.html             | 2660 kB | Astrochemistry module.                                                                                              |
| chimorga.html          | 194 kB  | Organic chemistry module.                                                                                           |
| proxy/spectra-proxy.js | 10 kB   | Optional proxy (Cloudflare Worker) for the AI calls.                                                                |

## Digests

| File                   | SHA-256                                                          |
|------------------------|------------------------------------------------------------------|
| index.html             | 967111a607a00049217685c40b9db62707fb8c0105a4e49c1cf466cb62597451 |
| bsi-ai-hub.js          | 985b9fa3f6a3d330cc6f7e31456e16760d0a0d9482f760b04d83765a013f2476 |
| bsi-spettri.js         | be757f15a70f884e3374d401596e2dfd18bcc852cafb4584920e2b2d24eb3c9  |
| bsi-cheminfo.js        | 59a3111b3b9436add5376fde1be6f90ded3dc08c095cebd04da50c7f8268bbc  |
| bsi-prettsch.js        | 27f727240363579fba3dba50ab3b567fbcd26a8a9a06c0e43968b195c479db3  |
| bsi-nmr.js             | c090d614ed5d7ed1fb958f6efec1fc4021cdc7b0f4fd6e1ca89bba167528d259 |
| bsi-moldraw.js         | c2c2a4cc258215a28b64ac3b06ed60befa343db930964be4a7f3360064c1f30e |
| bsi-spettrolettore.js  | dbdc79aa8e6fc14d26ae9633f15598eb7121efd6e1cf0118e5b9ff99681c567c |
| bsi-mol3d.js           | beb85c594db8e21bd1653ac6923989375a729d163b2749c6259ee48e05501ca9 |
| bsi-lingue.js          | 588beac68e5ea9fcfdc430a3b56290eb1df509a41c507aa24ba8c146a809431  |
| bsi-molingue.js        | e45fec1ee20e3e7b9a7bea3d6e20aa4e33e74ec43578bfa7bfb94646b7bc53fb |
| bsi-pannelli-lingua.js | 4af94d70ef104cf21be18ff7fbc7a2248c56e376e2ae48e8cac83878f4554f93 |
| sw.js                  | 83edcea62f21e8a315560127f7f798fa1256561c25998f873337f011f16fefe4 |
| rdkit_lab.html         | 066a42092b829df2fa306b691b9d162b6d44edfdb5317ce4cc8a276bc34c80ad |
| astro.html             | 3376b2b7f26ed483f2d698c6692b31f21c9f245f87d1fb1884216fb8b1da489d |

| File                   | SHA-256                                                          |
|------------------------|------------------------------------------------------------------|
| chimorga.html          | 047934abc73a79fbaf06ed7036b6e89a913ebdf5354fc52cecdbee7e8f94f51d |
| proxy/spectra-proxy.js | 808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962 |

### 3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

| Package         | Version | Licence    |
|-----------------|---------|------------|
| playwright-core | 1.62.1  | Apache-2.0 |

### 4. Runtime dependencies

**None.** The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

| Service                      | Purpose                                        | If unreachable                                   |
|------------------------------|------------------------------------------------|--------------------------------------------------|
| PubChem PUG-REST             | IUPAC name, CAS, GHS pictograms, 3D conformers | the app uses local data and says so              |
| NASA / ESA                   | astronomical images and data                   | the module shows the tabulated data              |
| AI providers (user's choice) | assistant answers                              | the assistant reports it and offers alternatives |

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage`, or on the user's own proxy (see `proxy/`).

### 5. Licence compatibility

All licences of the distributed components (MIT, BSD-3-Clause, CC0, Public Domain) are **permissive** and mutually compatible. No component is subject to a strong copyleft licence (GPL/AGPL): there is no redistribution constraint on the code written for this project.

The remaining obligations are of **attribution**: the licence notices of the upstream projects are reproduced in `THIRD_PARTY_NOTICES.md`.

**The application's licence is distinct from those of its components.** The code written for BioSpecInfo is **proprietary** — *All rights reserved* (see [LICENSE](#)): readable and open to assessment, but copying, reuse and commercial use require the Author's written permission. The permissive licences listed above cover **only** the bundled third-party libraries, and do not extend to the application.

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